

Azure Service Bus – Filters and Differences Between Filter Types

1. What is a Filter in Azure Service Bus?

A **filter** is a rule associated with a Service Bus topic subscription that determines **which messages the subscription should receive**.

In simple words:

A filter decides whether a message should be delivered to a particular subscription.

Example

Suppose we have:

```
Subscription: S1  
  
Filter:  
PaymentStatus = 'Completed'
```

S1 will receive messages where the relevant message property matches:

```
PaymentStatus = Completed
```

For example:

```
Message 1:  
PaymentStatus = Completed  
→ S1 receives   
  
Message 2:  
PaymentStatus = Pending  
→ S1 doesn't receive 
```

2. Where Can Filters Be Applied?

Filters can evaluate **message properties**.

These properties can include:

- System properties
- Application/User properties

Application/User Property Example

```
PaymentStatus = Completed
```

System Property Example

Service Bus messages also have system properties such as:

```
MessageId  
CorrelationId  
Subject
```

The important point is:

Service Bus filters work with message properties, not by directly searching inside the message body.

3. Message Body vs Message Properties

Suppose the message body is:

```
{  
  "OrderId": 101,  
  "PaymentStatus": "Completed"  
}
```

And the message also has an application property:

```
PaymentStatus = Completed
```

A filter can evaluate the **message property**:

```
PaymentStatus = 'Completed'
```

It does not simply search the JSON body for:

```
"PaymentStatus": "Completed"
```

Therefore, remember:

```
Message Body
  ↓
Actual business data

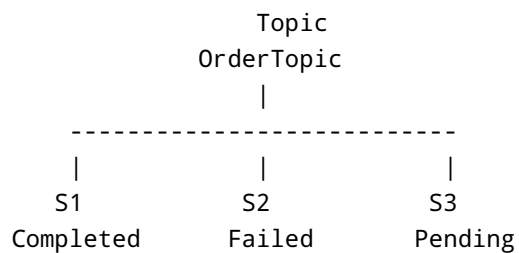
Message Properties
  ↓
Metadata used by filters
```

4. Why Do We Need Filters?

A topic can have multiple subscriptions.

Different subscriptions may need different messages.

For example:



Filters allow each subscription to receive only the messages it is interested in.

5. Types of Filters

The important filter types discussed here are:

1. **Boolean / True-False Filter**
2. **SQL Filter**
3. **Correlation Filter**

Let's understand each one.

6. Boolean / True-False Filter

A Boolean filter is a filter that always evaluates to either:

TRUE

or:

FALSE

True Filter

Example:

$1 = 1$

This is always true.

Therefore:

True Filter
↓
All messages are accepted

The subscription receives all messages.

False Filter

Example:

$1 = 0$

This is always false.

Therefore:

```
False Filter
↓
No messages are accepted
```

The subscription receives no messages.

7. Example of Boolean Filter

Suppose:

```
S1 → True Filter
S2 → False Filter
```

A message is sent to the topic.

Result:

```
S1 → TRUE → Message received ✓
S2 → FALSE → Message not received ✗
```

Easy Memory Trick

```
1 = 1
↓
TRUE
↓
Receive ALL

1 = 0
↓
FALSE
↓
Receive NONE
```

8. SQL Filter

A **SQL Filter** allows us to specify SQL-like expressions to determine whether a message should be delivered.

For example:

```
City = 'Mumbai'
```

This means:

Deliver the message if the `City` message property has the value `Mumbai`.

9. SQL Filter Supports More Complex Conditions

SQL filters can use operators and expressions such as:

```
=  
<>  
>  
<  
>=  
<=  
AND  
OR  
LIKE  
IN
```

For example:

Equality

```
City = 'Mumbai'
```

Greater Than

```
Amount > 10000
```

AND

```
City = 'Mumbai' AND Amount > 10000
```

OR

```
City = 'Mumbai' OR City = 'Delhi'
```

LIKE

```
City LIKE 'Mum%'
```

IN

```
City IN ('Mumbai', 'Delhi', 'Bangalore')
```

This makes SQL filters suitable for more complex filtering requirements.

10. SQL Filter Example

Suppose subscription S1 has:

SQL Filter:

```
City = 'Mumbai'
```

Message 1:

```
City = Mumbai
```

Result:

```
Mumbai = Mumbai
```

↓

```
TRUE
```

↓

```
S1 receives message 
```

Message 2:

```
City = Bangalore
```

Result:

```
Bangalore = Mumbai
    ↓
    FALSE
    ↓
S1 doesn't receive message ❌
```

11. Correlation Filter

A **Correlation Filter** is designed for simple property matching.

The main condition supported is:

```
Equality (=)
```

For example:

```
City = Mumbai
```

The subscription receives messages where:

```
City == Mumbai
```

12. Why Use Correlation Filter?

Use a correlation filter when the filtering requirement is simple.

For example:

```
PaymentStatus = Completed
```

or:


```
City = Mumbai
```

You don't need a complex SQL expression.

The filter simply checks whether the message property matches the expected value.

13. Correlation Filter vs SQL Filter

A useful way to understand it:

```
Correlation Filter
  ↓
Simple property equality
  ↓
City = Mumbai
```

Whereas:

```
SQL Filter
  ↓
SQL-like expression
  ↓
City = 'Mumbai' AND Amount > 10000
```

Therefore:

Correlation filters are useful for simple equality-based filtering, while SQL filters are useful for more complex filtering expressions.

14. Performance Difference

Correlation filters are generally more efficient for simple equality-based matching because they are optimized for property-value correlation rather than evaluating a general SQL expression.

So when the requirement is simply:

```
Property = Value
```

a correlation filter is a good choice.

For example:

```
PaymentStatus = Completed
```

Use:

```
Correlation Filter
```

When the requirement is:

```
PaymentStatus = Completed  
AND  
Amount > 10000
```

a:

```
SQL Filter
```

is more appropriate.

15. Quick Comparison

Feature	Boolean / True-False	SQL Filter	Correlation Filter
Purpose	Accept all / reject all	Complex filtering	Simple property matching
True/False	Yes	Yes, based on expression	Yes, based on equality
Property matching	No	Yes	Yes
Equality	Not applicable	Yes	Yes
AND / OR	No	Yes	No
LIKE	No	Yes	No
IN	No	Yes	No

Feature	Boolean / True-False	SQL Filter	Correlation Filter
Greater/Less than	No	Yes	No
Best for	All or no messages	Complex conditions	Simple equality
Example	<code>1 = 1</code>	<code>City = 'Mumbai' AND Amount > 10000</code>	<code>City = Mumbai</code>

16. Simple Real-Time Example

Imagine an e-commerce application.

We have a topic:

```
OrderTopic
```

And three subscriptions:

```
CompletedOrders
HighValueOrders
MumbaiOrders
```

Subscription 1

```
CompletedOrders
```

Filter:

```
PaymentStatus = 'Completed'
```

Subscription 2

```
HighValueOrders
```

Filter:

Amount > 50000

Subscription 3

MumbaiOrders

Filter:

City = Mumbai

Suppose a message has:

PaymentStatus = Completed
Amount = 75000
City = Mumbai

The message can match all three subscriptions.

CompletedOrders

→ TRUE 

HighValueOrders

→ TRUE 

MumbaiOrders

→ TRUE 

Therefore, the message can be delivered to all three subscriptions.

17. Important Interview Concept

Question:

Do Service Bus filters work on the message body?

Answer:

No. Service Bus filters evaluate **message properties**, such as application properties and supported system properties, rather than searching through the message body.

For example, if we want to filter using:

```
PaymentStatus = Completed
```

we should provide it as a message property.

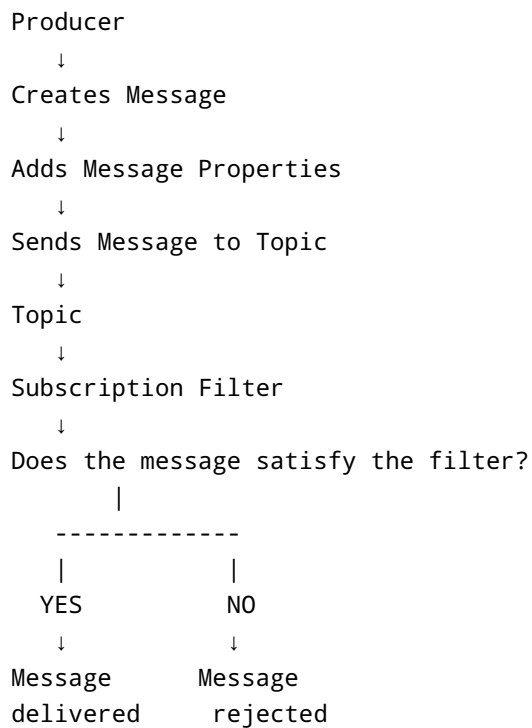
Example:

```
message.ApplicationProperties.Add(  
    "PaymentStatus",  
    "Completed");
```

Then the subscription filter can evaluate that property.

18. Filter Decision Flow

The overall process is:



19. How to Choose the Right Filter

Requirement 1: Receive all messages

Use:

```
True Filter  
1 = 1
```

Requirement 2: Receive no messages

Use:

```
False Filter  
1 = 0
```

Requirement 3: Simple equality

Example:

```
City = Mumbai
```

Use:

```
Correlation Filter
```

Requirement 4: Complex condition

Example:

```
City = 'Mumbai'  
AND Amount > 10000
```

Use:

```
SQL Filter
```

20. Interview Questions

Q1. What is a filter in Azure Service Bus?

A filter is a subscription-level rule that determines whether a message should be delivered to that subscription based on message properties.

Q2. What are the main types of filters?

The commonly discussed types are:

- Boolean/True-False filters
 - SQL filters
 - Correlation filters
-

Q3. What is a True Filter?

A True Filter always evaluates to `true`, so the subscription receives all messages.

Example:

```
1 = 1
```

Q4. What is a False Filter?

A False Filter always evaluates to `false`, so the subscription receives no messages.

Example:

```
1 = 0
```

Q5. What is a SQL Filter?

A SQL Filter uses SQL-like expressions to filter messages based on message properties.

Example:

```
City = 'Mumbai' AND Amount > 10000
```

Q6. What is a Correlation Filter?

A Correlation Filter is used for simple equality-based matching of message properties.

Example:

```
City = Mumbai
```

Q7. Which filter should you use for simple equality?

Use a **Correlation Filter**.

Example:

```
City = Mumbai
```

Q8. Which filter should you use for complex conditions?

Use a **SQL Filter**.

Example:

```
City = 'Mumbai'  
AND Amount > 10000
```

Q9. Do filters work on the message body?

No. Filters evaluate message properties, not the message body.

21. Quick Revision Notes

FILTER

↓

Decides which messages a subscription receives

Boolean Filter

1 = 1 → Receive ALL

1 = 0 → Receive NONE

SQL Filter

SQL-like expressions

Example:

City = 'Mumbai'

AND Amount > 10000

Supports conditions such as:

=

<>

>

<

>=

<=

AND

OR

LIKE

IN

Correlation Filter

Simple equality matching

Example:

City = Mumbai

Most Important Difference

Boolean

→ All or Nothing

Correlation

→ Simple Equality

SQL

→ Complex Conditions

One-Line Interview Answer

Azure Service Bus filters control which messages a subscription receives based on message properties. True/False filters accept all or no messages, correlation filters are optimized for simple equality matching, and SQL filters support more complex SQL-like expressions.